C

EST II — MATH LEVEL 2 • UNIT 1.01

Complex Numbers, Modulus & Polar Form

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

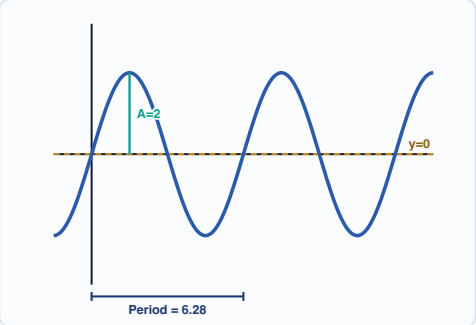
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 B	Q4 B	Q5 A
Q6 A	Q7 A	Q8 A	Q9 B	Q10 C
Q11 C	Q12 C	Q13 A	Q14 D	Q15 A
Q16 D	Q17 A	Q18 C	Q19 A	Q20 A
Q21 A	Q22 B	Q23 B	Q24 B	Q25 A

Complex Numbers, Modulus & Polar Form Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?



A $2 + 3i$ ✓

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

WORKED SOLUTION & VERIFICATION

To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is $2 + i$:

$$\frac{7+4i}{2-i} = \frac{(7+4i)(2+i)}{(2-i)(2+i)}$$

Expand the numerator using the distributive property (FOIL):

$$(7+4i)(2+i) = 14 + 7i + 8i + 4i^2$$

Since $i^2 = -1$:

$$14 + 15i + 4(-1) = 14 - 4 + 15i = 10 + 15i$$

Expand the denominator using the difference of squares:

$$(2-i)(2+i) = 2^2 - i^2 = 4 - (-1) = 5$$

Now, divide each term in the numerator by the denominator:

$$\frac{10+15i}{5} = \frac{10}{5} + \frac{15i}{5} = 2 + 3i$$

Common Misconceptions:

- Choice B occurs from an arithmetic sign error in the imaginary term.
- Choice C occurs if the student mistakenly calculates the denominator as $2^2 - 1^2 = 3$ by forgetting that $-i^2 = -(-1) = +1$.
- Choice D occurs if the student takes $i^2 = 1$, yielding $14 + 4 = 18$ in the numerator.

2 MCQ

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

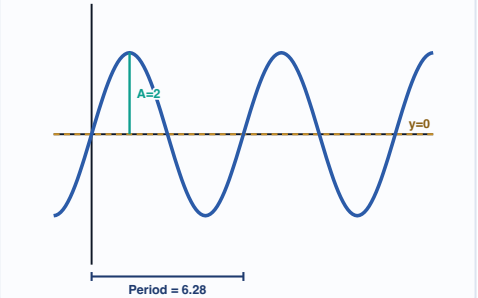
- **A** 5 ✓
- **B** 7
- **C** 12
- **D** 25

WORKED SOLUTION & VERIFICATION

The magnitude of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the magnitude is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

3 MCQ

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.



- **A** 5
- **B** 6 ✓
- **C** 7
- **D** 8

WORKED SOLUTION & VERIFICATION

Let $w = (a + bi) + (3 - 2i) = (a + 3) + (b - 2)i$. We are given that $|w|^2 = 25$, so $(a + 3)^2 + (b - 2)^2 = 25$. Since w lies in the first quadrant, $a + 3 > 0$ and $b - 2 > 0$. We need to find integer or simple fractional solutions for $(a + 3)$ and $(b - 2)$ that sum of squares is 25. The possible pairs of squares are $(9, 16)$ and $(16, 9)$.

Case 1: $(a + 3)^2 = 9$ and $(b - 2)^2 = 16$. This gives $a + 3 = \pm 3$ and $b - 2 = \pm 4$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 3$ and $b - 2 = 4$. This leads to $a = 0$ and $b = 6$. However, if $a = 0$, then $w = 3 + 4i$, and $a + b = 0 + 6 = 6$.

Case 2: $(a + 3)^2 = 16$ and $(b - 2)^2 = 9$. This gives $a + 3 = \pm 4$ and $b - 2 = \pm 3$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 4$ and $b - 2 = 3$. This leads to $a = 1$ and $b = 5$. In this case, $w = 4 + 3i$, which is in the first quadrant. We want to find $a + b$. So, $a + b = 1 + 5 = 6$.

Both cases yield $a + b = 6$.

4 MCQ

Let $z_1 = 2 + 3i$ and $z_2 = 1 - i$. Consider the complex number w such that $w = \frac{z_1^2}{z_2}$. If $w = a + bi$, where a and b are real numbers, find the value of b.

A $\frac{17}{2}$

B $-\frac{17}{2}$ ✓

C $\frac{11}{2}$

D $-\frac{11}{2}$

WORKED SOLUTION & VERIFICATION

First, calculate z_1^2: $z_1^2 = (2 + 3i)^2 = 2^2 + 2(2)(3i) + (3i)^2 = 4 + 12i + 9i^2 = 4 + 12i - 9 = -5 + 12i$.

Next, divide z_1^2 by z_2: $w = \frac{-5+12i}{1-i}$.

To simplify this division, multiply the numerator and the denominator by the conjugate of the denominator, which is $1 + i$:

$w = \frac{-5+12i}{1-i} \times \frac{1+i}{1+i} = \frac{(-5+12i)(1+i)}{(1-i)(1+i)}$.

The numerator is: $(-5)(1) + (-5)(i) + (12i)(1) + (12i)(i) = -5 - 5i + 12i + 12i^2 = -5 + 7i - 12 = -17 + 7i$.

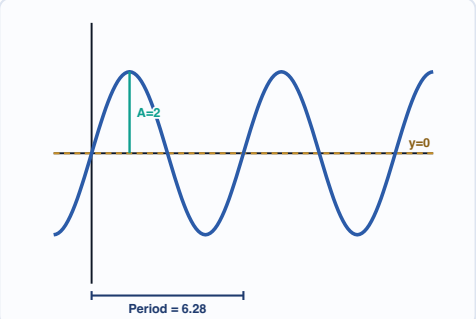
The denominator is: $(1)^2 - (i)^2 = 1 - (-1) = 1 + 1 = 2$.

So, $w = \frac{-17+7i}{2} = -\frac{17}{2} + \frac{7}{2}i$.

Given that $w = a + bi$, we have $a = -\frac{17}{2}$ and $b = \frac{7}{2}$. The question asks for the value of b. Therefore, $b = \frac{7}{2}$.

5 MCQ

Let $z = 3 - 4i$. What is the real part of z?



A 3 ✓

B -4

C 5

D 7

WORKED SOLUTION & VERIFICATION

A complex number is generally expressed in the form $a + bi$, where a is the real part and b is the imaginary part. In the given complex number $z = 3 - 4i$, the real part is the term that does not have i multiplied by it. Therefore, the real part of z is 3.

6 MCQ

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.

- **A** 6 ✓
- **B** 8
- **C** 10
- **D** 12

WORKED SOLUTION & VERIFICATION

The quadratic equation is $x^2 - 4x + 5 = 0$. By Vieta's formulas, the sum of the roots is $z_1 + z_2 = -(-4)/1 = 4$ and the product of the roots is $z_1 z_2 = 5/1 = 5$.
We are interested in $w = z_1^2 + z_2^2$. We can rewrite this as $w = (z_1 + z_2)^2 - 2z_1 z_2$.
Substituting the values from Vieta's formulas:
$w = (4)^2 - 2(5)$
$w = 16 - 10$
$w = 6$.
So, $w = 6$. When expressed in the form $a + bi$, we have $a = 6$ and $b = 0$.
The question asks for $a - b$.
$a - b = 6 - 0 = 6$.

7 MCQ

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

- **A** 5 ✓
- **B** 7
- **C** 12
- **D** 25

WORKED SOLUTION & VERIFICATION

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$.
Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

8 MCQ

What is i^2?

- **A** -1 ✓
- **B** 1
- **C** i
- **D** $-i$

WORKED SOLUTION & VERIFICATION

Applying Complex Numbers principles, By definition $i^2 = -1$. Therefore, the verified correct option is -1.

9 MCQ

Express in simplest algebraic form $(9 + i) + (-7 + i)$.

- **A** 3+2i
- **B** 2+2i ✓
- **C** 2+4i
- **D** 16

WORKED SOLUTION & VERIFICATION

Applying Complex Numbers principles, $(9 + -7) + (1 + 1)i = 2 + 2i$. Therefore, the verified correct option is 2+2i.

10 MCQ

Express in simplest algebraic form $(2-9i)-(1+7i)$.

A 2-16i	B 1-14i
C **1-16i** ✓	D 3-2i

WORKED SOLUTION & VERIFICATION

Applying Complex Numbers principles, $(2-1)+(-9-7)i=1-16i$. Therefore, the verified correct option is 1-16i.

11 MCQ

Let the complex number $x=24+14i$ have complex conjugate $\bar{x}$ on the Argand diagram. What is the squared modulus $f=x\bar{x}=|x|^2$?

A 796	B 758
C **772** ✓	D 634

WORKED SOLUTION & VERIFICATION

Multiplying x by $\bar{x}$ gives $24^2+14^2=772$.

12 MCQ

Express in simplest algebraic form i^{26}.

A 1	B i
C **-1** ✓	D -i

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $26 \bmod 4=2$, so $i^{26}=-1$.

13 MCQ

Rewrite in fully simplified form i^{21}.

A **i** ✓	B -i
C 1	D -1

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $21 \bmod 4=1$, so $i^{21}=i$.

14 MCQ

Reduce to its simplest form i^7.

A 1	B i
C -1	D **-i** ✓

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $7 \bmod 4=3$, so $i^7=-i$.

15 MCQ

Express in simplest algebraic form i^{34}.

- **A** -1 ✓
- **B** 1
- **C** i
- **D** -i

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $34 \bmod 4 = 2$, so $i^{34} = -1$.

16 MCQ

Rewrite in fully simplified form i^{3}.

- **A** 1
- **B** i
- **C** -1
- **D** -i ✓

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $3 \bmod 4 = 3$, so $i^{3} = -i$.

17 MCQ

Reduce to its simplest form i^{9}.

- **A** i ✓
- **B** 1
- **C** -1
- **D** -i

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $9 \bmod 4 = 1$, so $i^{9} = i$.

18 MCQ

Express in simplest algebraic form i^{8}.

- **A** i
- **B** -1
- **C** 1 ✓
- **D** -i

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $8 \bmod 4 = 0$, so $i^{8} = 1$.

19 MCQ

Rewrite in fully simplified form i^{30}.

- **A** -1 ✓
- **B** 1
- **C** i
- **D** -i

WORKED SOLUTION & VERIFICATION

Powers of i cycle with period 4; $30 \bmod 4 = 2$, so $i^{30} = -1$.

20 MCQ

Express in simplest algebraic form $\frac{-5-5i}{5+5i}$.

- **A** -1 ✓
- B -i
- C 0
- D 1

WORKED SOLUTION & VERIFICATION

Multiply by the conjugate $5-5i$ over itself: numerator $=-50$, denominator $=|5+5i|^2=50$; result $=-1$.

21 MCQ

Express in simplest algebraic form $\frac{1+7i}{2-i}$.

- **A** -1+3i ✓
- B 3-i
- C 3i
- D 1+3i

WORKED SOLUTION & VERIFICATION

Multiply by the conjugate $2+i$ over itself: numerator $=-5+15i$, denominator $=|2-i|^2=5$; result $=-1+3i$.

22 MCQ

Express in simplest algebraic form $\frac{-2+6i}{2-6i}$.

- A -i
- **B** -1 ✓
- C 0
- D 1

WORKED SOLUTION & VERIFICATION

Multiply by the conjugate $2+6i$ over itself: numerator $=-40$, denominator $=|2-6i|^2=40$; result $=-1$.

23 MCQ

Rewrite in fully simplified form $\frac{-7-9i}{1+2i}$.

- A 1-5i
- **B** -5+i ✓
- C -4+i
- D 5+i

WORKED SOLUTION & VERIFICATION

Multiply by the conjugate $1-2i$ over itself: numerator $=-25+5i$, denominator $=|1+2i|^2=5$; result $=-5+i$.

24 MCQ

Simplify $(5 + 2i) + (3 - 4i)$.

- A $8 + 2i$
- B $8 - 2i$ ✓
- C $2 - 2i$
- D $8 - 6i$

WORKED SOLUTION & VERIFICATION

Applying Complex Numbers principles, $(5 + 3) + (2 - 4)i = 8 - 2i$. Therefore, the verified correct option is $8 - 2i$.

25 MCQ

Simplify $(7 - 3i) - (2 - i)$.

- A $5 - 2i$ ✓
- B $5 - 4i$
- C $9 - 2i$
- D $5 + 2i$

WORKED SOLUTION & VERIFICATION

Applying Complex Numbers principles, $(7 - 2) + (-3 - (-1))i = 5 - 2i$. Therefore, the verified correct option is $5 - 2i$.